

Quiz (Carolina Reaper)

- Q1.** A shipping company has $3 \cdot 10^2$ trucks and $4 \cdot 10^2$ vans. What is the ratio of trucks to vans?
(A) 3:4
(B) 3:2
(C) 2:3
(D) 4:3
(E) 1:1
- Q2.** If 15 packages are delivered in 3 hours, what is the rate of delivery per hour?
(A) 5 packages/hour
(B) 3 packages/hour
(C) 1 package/hour
(D) 2 packages/hour
(E) 4 packages/hour
- Q3.** A traffic signal has a ratio of red to green of 2 : 3. If the signal is red for 6 minutes, how many minutes is it green?
(A) 6 minutes
(B) 8 minutes
(C) 9 minutes
(D) 7 minutes
(E) 5 minutes
- Q4.** A logistics company has 12 warehouses and 8 distribution centers. What is the ratio of warehouses to distribution centers?
(A) 3:2
(B) 2:1
(C) 1:1
(D) 3:1
(E) 2:3
- Q5.** If a delivery truck travels 250 miles in 5 hours, what is its average rate of travel?
(A) 50 miles/hour
(B) 25 miles/hour
(C) 40 miles/hour
(D) 60 miles/hour
(E) 30 miles/hour
- Q6.** A transportation company has a ratio of cars to buses of 5 : 2. If there are 10 buses, how many cars are there?
(A) 20 cars
(B) 25 cars
(C) 12 cars
(D) 15 cars
(E) 10 cars
- Q7.** If 18 boxes are packed into 3 containers, what is the ratio of boxes to containers?
(A) 6:1
(B) 3:1
(C) 2:1
(D) 1:1
(E) 9:1
- Q8.** A shipping container can hold 24 crates. If $\frac{2}{3}$ of the container is filled, how many crates are in the container?
(A) 16 crates
(B) 12 crates
(C) 8 crates
(D) 20 crates
(E) 18 crates

Open Ended Questions (FRQ)

- Q9.** A delivery company has 5 drivers and 8 trucks. If the ratio of drivers to trucks is 3 : 4, how many more trucks should be purchased to meet this ratio? Show your work and explain your reasoning.
- Q10.** A logistics company has a ratio of 2 : 5 for the number of packages delivered to the number of packages received. If 150 packages are received, how many packages should be delivered to maintain this ratio? Use proportions to solve the problem and explain your answer.

Chef's Tips

- Tip 1: Encourage students to use proportions to solve ratio problems
- Tip 2: Remind students to simplify their answers to the simplest form
- Tip 3: Emphasize the importance of labeling units in rate and ratio problems